ETE/EVEN/2024-25/403/Set 2

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JIS UNIVERSITY

2025

END TERM EXAMINATIONS, AY 2024 - 2025(EVEN)

YCS4002 - FORMAL LANGUAGE AND AUTOMATA THEORY

Time: 2 Hrs

Maximum Marks : 50

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any Ten (10x1=10 Marks)

1. The ratio of number of input to the number of output in a mealy machine can be given as: **(1)** CO1 BL2 PO1

   (a) 1 (b) n: n+1
   (c) n+1: n (d) None of the mentioned

2. Which of the following statements about finite automata is true? **(1)** CO1 BL2 PO1

   (a) Every DFA has an equivalent NFA (b) Every NFA has an equivalent DFA
   (c) NFAs can recognize some languages that DFAs cannot (d) Both (a) and (b)

3. Which of the following grammars defines a regular language? **(1)** CO3 BL1 PO2

   (a) $A \rightarrow aA \mid b$ (b) $A \rightarrow AB \mid a$
   (c) $A \rightarrow aBb \mid \varepsilon$ (d) $A \rightarrow aAb \mid \varepsilon$

4. For a give Moore Machine, Given Input='101010', thus the output would be of length: **(1)** CO1 BL2 PO1

   (a) |Input|+1 (b) |Input|
   (c) |Input|-1 (d) Cannot be predicted

5. The language of all strings with an even number of a's over {a, b} is: **(1)** CO1 BL1 PO1

   (a) Regular (b) Context-free but not regular

(c) Recursive but not context-free (d) Not recursive

6. The language recognized by a non-deterministic finite automaton is always: (1) CO1 BL2 PO1

(a) Regular (b) Context-free
(c) Context-sensitive (d) Recursive

7. The Myhill-Nerode theorem is used to: (1) CO3 BL3 PO1

(a) Prove that a language is context-free (b) Minimize the number of states in a DFA
(c) Find the time complexity of an algorithm (d) Convert a DFA to an NFA

8. Which of the following cannot be recognized by a finite automaton? (1) CO1 BL4 PO1

(a) Strings containing equal numbers of 0s and 1s (b) Strings ending with '01'
(c) Strings where every 0 is followed by at least one 1 (d) Strings containing at least one '1'

9. Which of the technique can be used to prove that a language is non regular? (1) CO3 BL2 PO1

(a) Pumping Lemma (b) Arden's theorem
(c) Ogden's Lemma (d) Thompson's rule

10. A Language for which no DFA exist is a________ (1) CO1 BL2 PO1

(a) Regular Language (b) Non-Regular Language
(c) May be Regular (d) Cannot be said

11. Which one of the following languages over the alphabet {0,1} is described by the regular expression: (0+1)*0(0+1)*0(0+1)*? (1) CO2 BL3 PO1

(a) The set of all strings containing the substring 00 (b) The set of all strings containing at most two 0's.
(c) The set of all strings containing at least two 0's (d) The set of all strings that begin and end with either 0 or 1.

12. The language of all strings over {0,1} that contain an even number of 0s is: (1) CO1 BL2 PO1

(a) Not regular (b) Regular and can be represented by a DFA
(c) Regular but cannot be represented by a DFA (d) Context-free but not regular

Part B
Answer any Two (2x5=10 Marks)

13. Define Deterministic Finite Automata (DFA) and design a DFA to accept all strings over {0,1} that end with '01'. (5) CO1 BL3 PO1

Evaluate the given grammar and determine a reduced grammar equivalent to: (5) CO5 BL5 PO1
G = ({S, A, B, C}, {a, b}, P, S)
with production rules:
S → aAa
A → bBB
B → ab
C → aB

Define NFA. Construct an NFA for the language that accepts all the binary strings which contains 01 as substring. (5) CO1 BL3 PO1

State the Pumping Lemma. Prove that L = $\{a^{i^2} \mid \text{where } i \geq 1\}$ is not regular using Pumping Lemma. (5) CO2 BL2 PO1

Part C
Answer any Three (3x10=30 Marks)

7. Minimize the following DFA: (10) CO3 BL4 PO1

States	Input: 0	Input: 1
->q0	q1	q5
q1	q6	q2
* q2	q0	q2
q3	q2	q6
q4	q7	q5
q5	q2	q6
q6	q6	q4
q7	q6	q2

18. Answer the following:

(a) Define grammar with example. Classify Grammar with examples. (5) CO4 BL2 PO1

(b) Show that the following Grammar is ambiguous: (5) CO5 BL2 PO1

S → aB | ab

A → aAB | a

B → ABb | b

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19. Design a Non-deterministic Finite Automaton (NFA) for the regular expression (aa + bb)(ab + ba)(aa + bb). Then, develop its equivalent Deterministic Finite Automaton (DFA) by applying the subset construction method. Demonstrate the complete transformation process. (10) CO2 BL6 PO1

20. Minimize the following DFA: (10) CO3 BL3 PO1

State	0	1
->A	B	C
B	D	E
C	F	G
D	D	E
E	F	G
*F	D	E
G	F	G

21. Answer the following:

(a) Obtain the regular expression from the following DFA using Arden's theorem. (5) CO1 BL1 PO1

State	a	b
-> * q1	q2	q3
q2	q4	q1
q3	q1	q4
q4	q4	q4

(b) Design a DFA for the language which will accept all the starings having even number of a and odd number of b. ∑ = {a, b}*. (5) CO1 BL3 PO1

CO Definitions:

1) Explain the basic properties of formal languages and grammars.
2) Understand the tools for recognizing different formal languages.
3) Differentiate between regular, context-free, and recursively enumerable languages.
4) Apply the theory of computation and computational models, including decidability and intractability.
5) Analyze complex problems and automata to find solutions using the theory of computation and computational models, including decidability and intractability.